

NYCU 2026 Spring

Cloud Native Development and Best Practice

授課單位：台積電企業資訊技術 (IT)

A graphic showing a glowing blue cloud icon on a dark, textured surface resembling a circuit board. The text "CLOUD NATIVE Development Best Practice" is overlaid in white.

CLOUD NATIVE
Development Best Practice

Cloud Native Development and Best Practice

的第一堂課

Greeting

曾建超、黃俊龍教授

Embracing Cloud Native

台積電 / William Yeh

Course Introduction.....

台積電 / 教學團隊

Q&A / 加簽

Cloud Native Development and Best Practice

的第一堂課

Greeting

曾建超、黃俊龍教授

Embracing Cloud Native

台積電 / William Yeh

Course Introduction.....

台積電 / 教學團隊

Q&A / 加簽



Unleash Innovation

Embracing Cloud Native

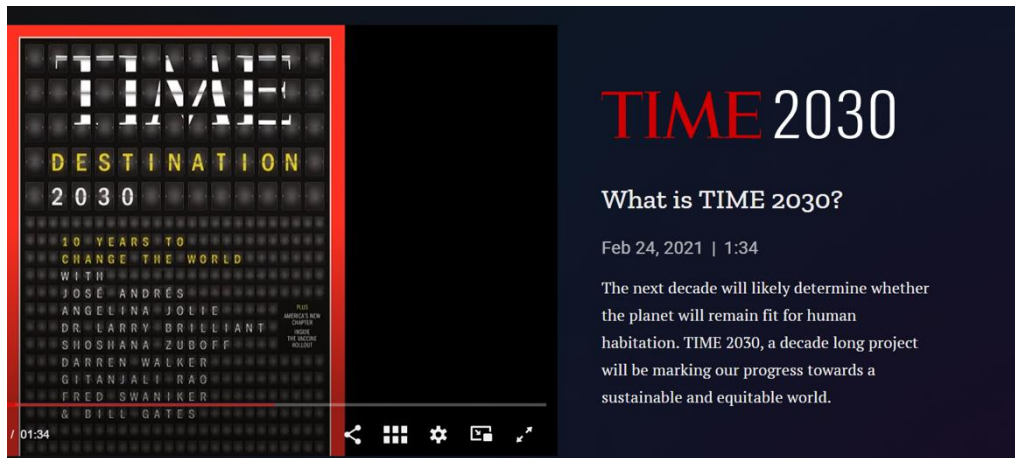
William Yeh / TSID

Why TSMC needs Cloud Native

— a Digital Transformation journey

2030 – 10 years to change the world

- Covid-19 has reshaped people's lifestyles. Daily activities, working models and consumption patterns has become **heavily reliant on technology**.
- **TSMC** is facing unprecedented rapid expansion with **increased production capacity** and **global factory establishments**.



邁向2030年，科技發展對人類的重要影響
愈趨明顯

INNOVATION

Inside the Taiwan
Firm That Makes the
World's Tech Run



台灣是全世界最重要的半導體生產基地

用 IT 重新定義半導體的製造和生產 (2021)

■ 「台積電一直用 IT 重新定義半導體的製造和生產」

- ✓ 30年台積電 IT 發展，從 自動製造、智能製造 到 數位轉型
- ✓ AI 應用、元宇宙、物聯網、5G/6G 等浪潮帶來變革、機會與挑戰
- ✓ 先進製程軟硬體整合高，複雜到人力難以處理，需要 IT 緊密支援
- ✓ 全球擴廠，全球化營運是台積電 IT 新挑戰

■ 台積電數位轉型

- ✓ 重塑 IT 組織
- ✓ 改造 IT 架構
- ✓ 擁抱 Cloud Native 技術

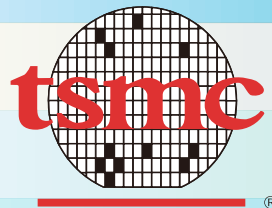


CIO, Chris

Treat IT as a Software Tech Company. Incorporate Silicon Valley Best Practices. How can we make a difference?

晶圓製造的第一，資訊技術 急起直追

- Software is Eating the World
- TSMC Approach - Scaling Digitally
- It Is a Race



FAANG

Scale

Speed

Service

Data-Driven

TSMC and IT's Digital Transformation

數位轉型 → 技術轉型、流程優化、人和文化的改變



數位轉型後：產品導向的團隊重構

智慧工廠、數位商務

- IT 是半導體前瞻製程相當重要的支援夥伴，每一個單位都會用到 IT 的系統跟平台
- 強化整合程度、AI化、穩定度，就能直接增進台積電的競爭優勢

晶圓廠自動化、良率分析、生產力、研發

製程良率
分析

智慧化
生產製造系統

數位化運營
平台

品質管理

整合製程
控制

高階運算
架構

廠房設施

機台自動化
與優化

現代數據與
安全平台

數位商務

CRM

客戶協作平台

SCMP

供應鏈規劃與管理

PLM

技術與產品管理

HCM

人力資源管理

ERP

財務與物流管理

Digi Work

數位員工工作
流程和服務

It's 2026.

Does Cloud Native still matter?

(Since we have AI...)

Boosting Cloud-Native Technology to Accelerate AI Innovation

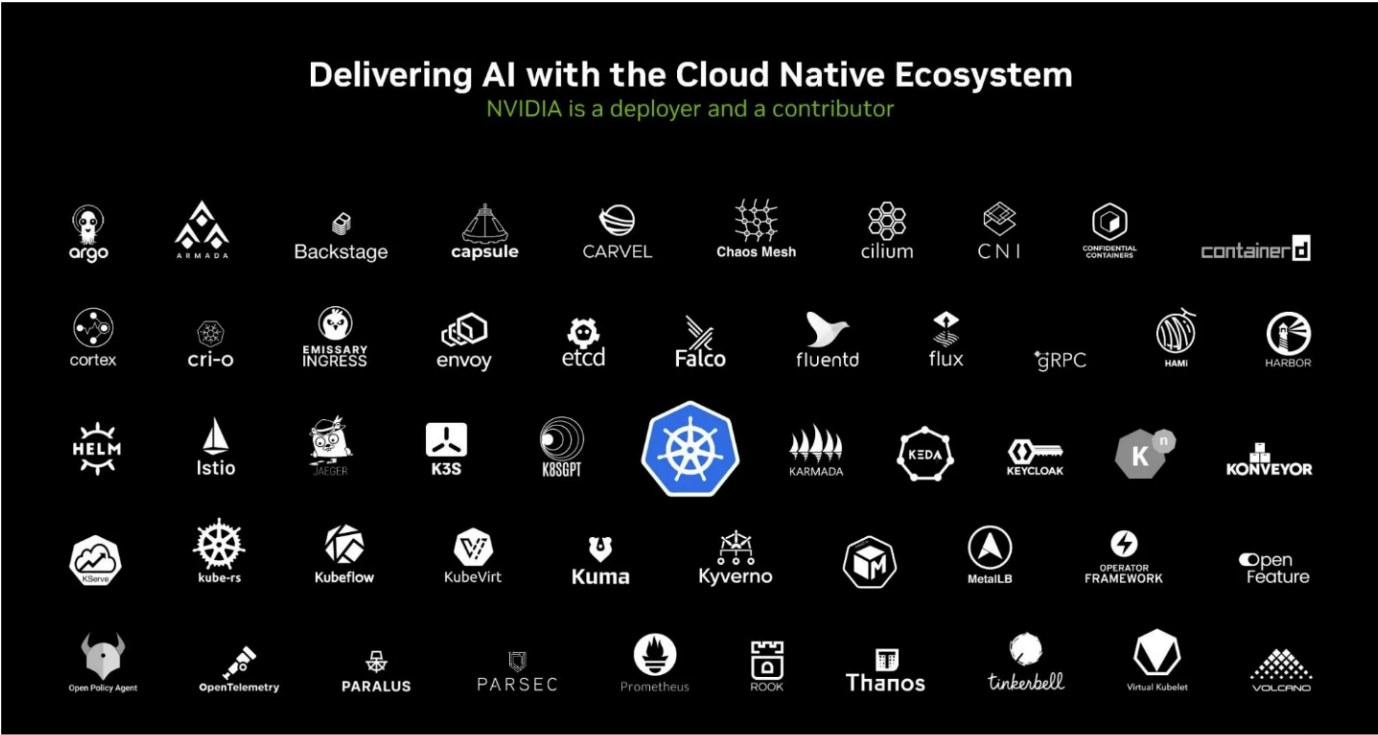


[Home](#) [AI](#) [Data Center](#) [Driving](#) [Gaming](#) [Pro Graphics](#) [Robotics](#) [Healthcare](#) [Startups](#) [AI Podcast](#) [NVIDIA Life](#)

Open for Development: NVIDIA Works With Cloud-Native Community to Advance AI and ML

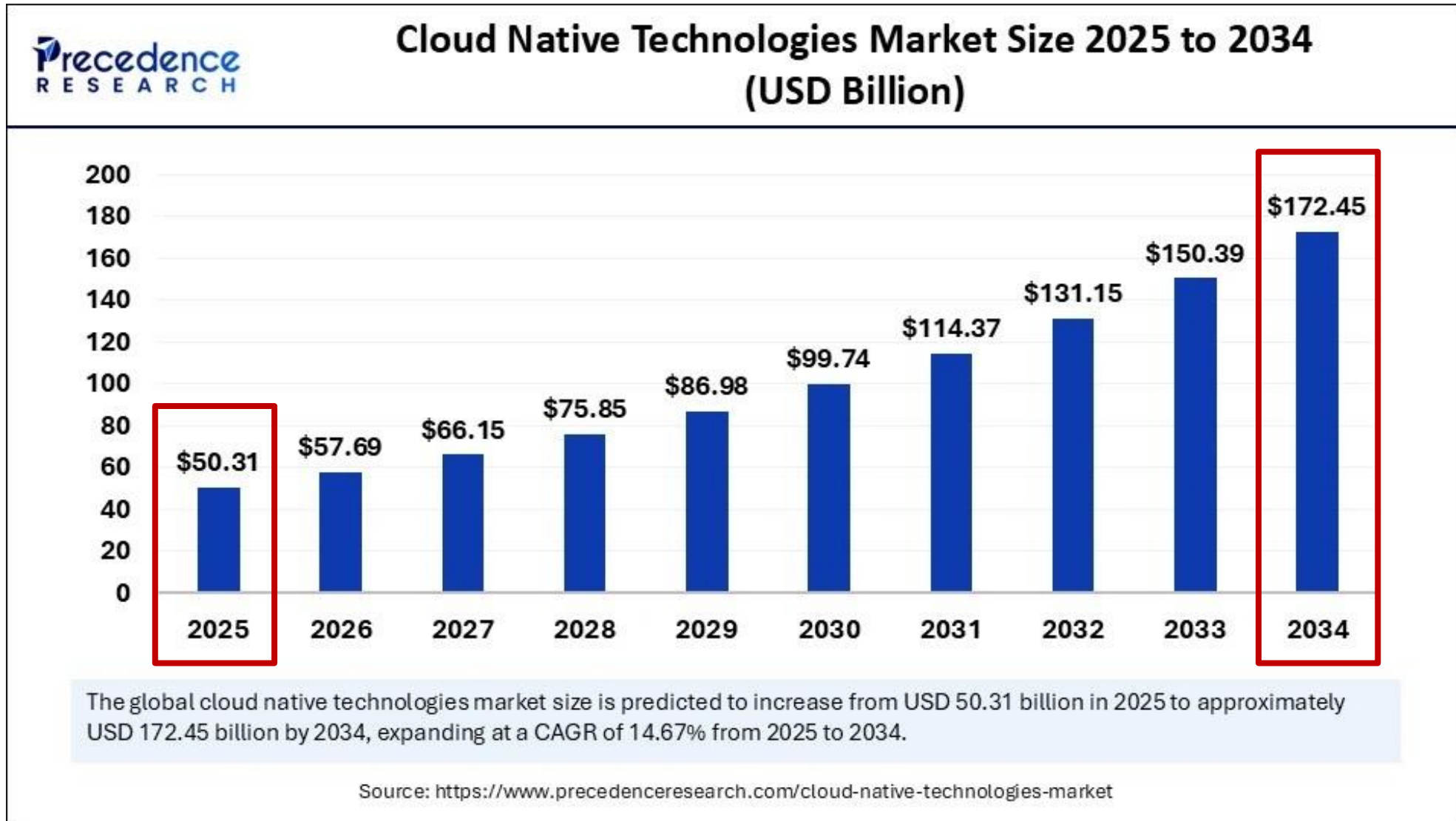
At KubeCon+CloudNativeCon, NVIDIA engineers and experts showcase open-source software contributions through a keynote, technical sessions, training and demos.

November 14, 2024 by [Ankit Patel](#)



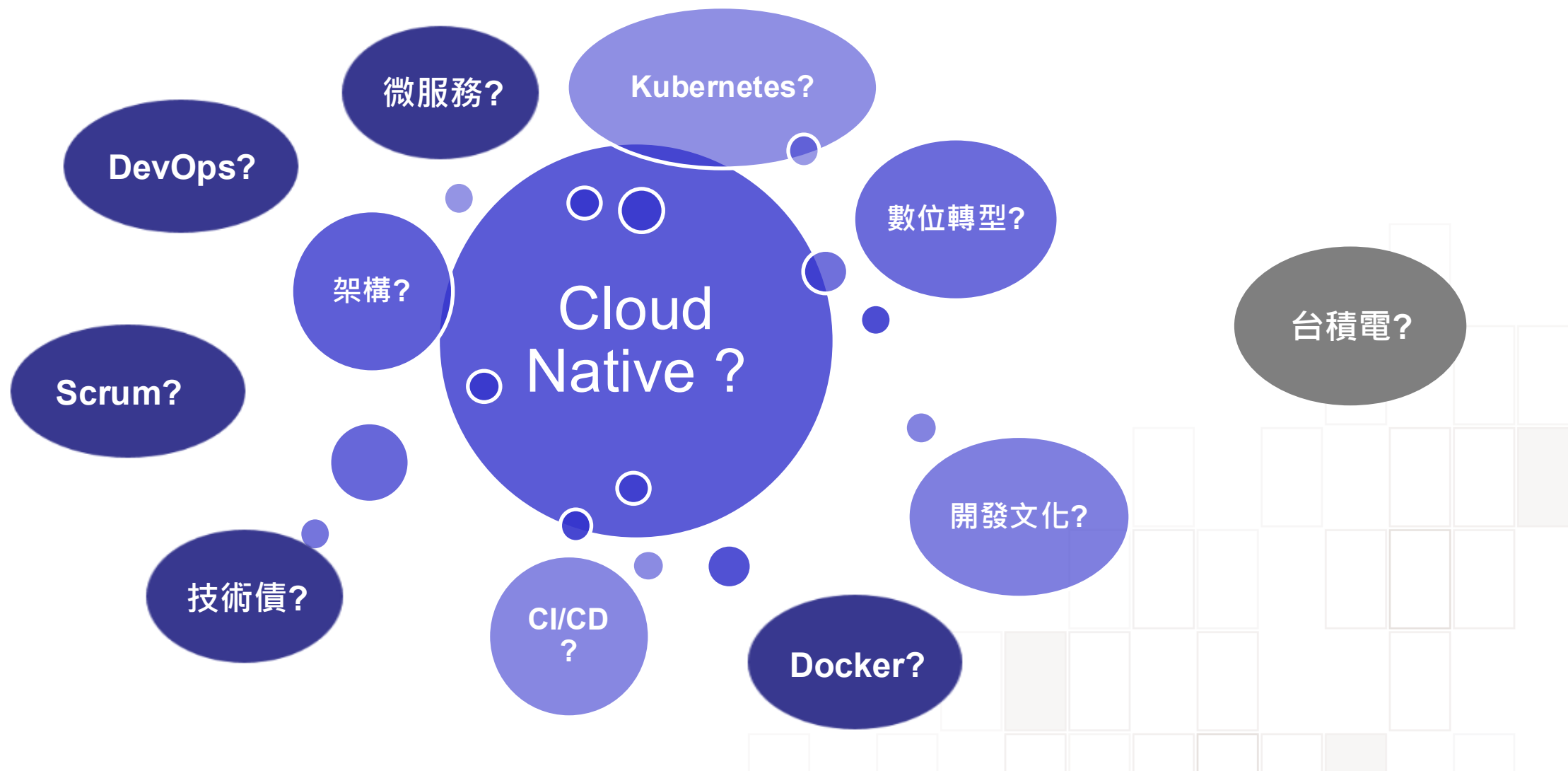
Open for Development: NVIDIA Works With Cloud-Native Community to Advance AI and ML (<https://blogs.nvidia.com/blog/open-source-cloud-native-ai-ml/>)

Cloud-Native Market: \$200B by 2030



What is Cloud Native?

Cloud Native (雲端原生) 是甚麼？



Cloud Native (雲端原生) 是甚麼？

Cloud Native Computing Foundation 的 官方定義：

- ✓ 雲端原生的實踐賦予組織能夠在**運算環境中（公有雲、私有雲、混合雲）**以**規模化、可程式化和可重複的方式開發、構建和部署工作負載**，以滿足其組織需求。
- ✓ 其特徵為**鬆耦合的系統**，以**安全、彈性、可管理、可持續和可觀察**的方式相互操作。
- ✓ 雲端原生技術和架構通常由**容器、服務網格、多租戶、微服務、不可變基礎設施、無伺服器**和**聲明式 API** 皆為此方法範例。

雲端原生的益處

- ✓ 搭配穩固的**自動化**，雲端原生的實踐讓組織以最少的勞動、清楚分離關注點，就能**頻繁、可預測地進行重大變更**。

CNCF關注點

- ✓ 雲端原生運算基金會（Cloud Native Computing Foundation）旨在透過培育和維持一個開放原始碼、廠商中立的專案生態系統，以推動此範式的採用。我們將最先進的模式民主化，使這些創新對每個人都可用。

Cloud Native Trail Map

- Recommended process for leveraging open-source software (marked blue: will introduce in class)

1. Containerization
2. CI/CD
3. Orchestration & application definition
4. Observability & analysis
5. Service proxy, discovery & mesh
6. Networking, policy & security
7. Distributed database & storage
8. Streaming & messaging
9. Container registry & runtime
10. Software distribution



CLOUD NATIVE TRAIL MAP

The Cloud Native Landscape [Landscape](#) has a large number of options. This Cloud Native Trail Map is a recommended process for leveraging open source, cloud native technologies. At each step, you can choose a vendor supported offering or do it yourself, and everything after step 10 is optional based on your circumstances.

HELP ALONG THE WAY

A. Training and Certification

Consider training offerings from CNCF and then take the exam to become a Certified Kubernetes Administrator or a Certified Kubernetes Application Developer [cnf.io/training](#)

B. Consulting Help

If you want assistance with Kubernetes and the surrounding ecosystem, consider leveraging a Kubernetes Certified Service Provider [cnf.io/kspp](#)

C. Join CNCF's End User Community

For companies that don't offer cloud native services externally [cnf.io/enduser](#)

WHAT IS CLOUD NATIVE?

Cloud native technologies empower organizations to build and run scalable applications in modern, dynamic environments such as public, private, and hybrid clouds. Containers, service meshes, microservices, immutable infrastructure, and declarative APIs exemplify this approach.

These techniques enable loosely coupled systems that are resilient, manageable, and observable. Combined with robust automation, they allow engineers to make high-impact changes frequently and predictably with minimal toil.

The Cloud Native Computing Foundation seeks to drive adoption of this paradigm by fostering and sustaining an ecosystem of open source, vendor neutral projects. We democratize state-of-the-art patterns to make these innovations accessible for everyone.

[l.cncf.io](#)

v20200501



1. CONTAINERIZATION

- Commonly done with Docker containers
- Any size application and dependencies (even PDP-T1 code running on an emulator) can be containerized
- Over time, you should aspire towards splitting suitable applications and writing future functionality as microservices

3. ORCHESTRATION & APPLICATION DEFINITION

- Kubernetes is the market leading orchestration solution
- You should select a Certified Kubernetes Distribution, Hosted Platform, or Installer: [cnf.io/ck](#)
- Helm Charts help you define, install, and upgrade even the most complex Kubernetes application



5. SERVICE PROXY, DISCOVERY, & MESH

- CoreDNS is a fast and flexible tool that is useful for service discovery
- Envoy and Linkerd each enable service mesh architectures
- They offer health checking, routing, and load balancing



7. DISTRIBUTED DATABASE & STORAGE

When you need more resiliency and scalability than you can get from a single database, Vitess is a good option for running MySQL at scale through sharding. Rook is a storage orchestrator that integrates a diverse set of storage solutions into Kubernetes. Serving as the "brain" of Kubernetes, etcd provides a reliable way to store data across a cluster of machines. TiKV is a high performance, distributed transactional key value store written in Rust.



9. CONTAINER REGISTRY & RUNTIME

Harbor is a registry that stores, signs, and scans content. You can use alternative container runtimes. The most common, both of which are OCI compliant, are containerd and CRI-O.



2. CI/CD

- Setup Continuous Integration/Continuous Delivery (CI/CD) so that changes to your source code automatically result in a new container being built, tested, and deployed to staging and eventually, perhaps, to production
- Setup automated rollouts, roll backs and testing
- Argo is a set of Kubernetes-native tools for deploying and running jobs, applications, workflows, and events using GitOps paradigms such as continuous and progressive delivery and ML ops



4. OBSERVABILITY & ANALYSIS

- Pick solutions for monitoring, logging and tracing
- Consider CNCF projects Prometheus for monitoring, Fluentd for logging and Jaeger for tracing
- For tracing, look for an Open Tracing-compatible implementation like Jaeger



6. NETWORKING, POLICY, & SECURITY

To enable more flexible networking, use a CNI-compliant network project like Calico, Flannel, or Weave Net. Open Policy Agent (OPA) is a general-purpose policy engine with uses ranging from authorization and admission control to data filtering. Falco is an anomaly detection engine for cloud native.



8. STREAMING & MESSAGING

When you need higher performance than JSON-RPC, consider using gRPC or NATS. gRPC is a universal RPC framework. NATS is a multi-modal messaging system that includes request/reply, pub/sub and load balanced queues. CloudEvents is a specification for describing event data in common ways.



10. SOFTWARE DISTRIBUTION

If you need to do secure software distribution, evaluate Notary, an implementation of The Update Framework.



雲端原生軟體的要素：Open-Source

技術棧使用範例

Language



React



python



Database



Micro Service



kubernetes



docker



argo



Prometheus



Grafana



elasticsearch



logstash



kibana

Message Broker



Other



Jira



from CNCF Cloud Native Landscape, 2026

<https://landscape.cncf.io/>

Benefits of Adopting Cloud-native Computing

- **Speed, agility and productivity:**
 - Organizations can harness the innovation and shorten the time to market by embracing DevOps & GitOps practices via CI/CD (continuous integration / continuous delivery) pipelines.
- **Scalability and resilience:**
 - Cloud-native applications can be automatically scaled to meet surge requests instantly by leveraging the scalability, elasticity, resiliency, and flexibility the cloud service provides.
- **Cost saving:**
 - Resources can be automatically scaled out / in fast as needed serving the dynamic demands without having to allocate fixed size of capacity so as to optimize the usage with lower TCO.
- **Vendor lock-in reduction:**
 - The methodology of building a cloud-native application allows an enterprise to run it on multiple clouds (public or on-premises) using an array of open-source software projects.
- **Attract and retain top talent, like You!**
 - The cloud-native technology enables developers to focus on value-added business and unleash people's innovations while repetitive operations are automated.

與 TSMC IT 一起進入 Cloud Native 世界

1. 藉由這門課讓同學提早接觸業界使用的雲原生技術，與產業無縫接軌
2. 介紹開源軟體的環境以及實作經驗，培養程式、開發能力